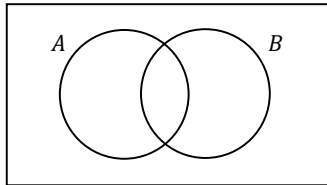


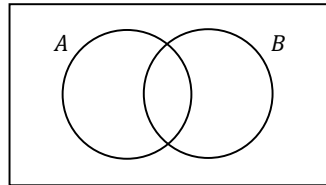
Discrete Mathematics: Foundations revision

- 1 The diagrams below illustrate a universal set $\mathcal{E}$ together with sets A and B . In each case shade the region indicated:

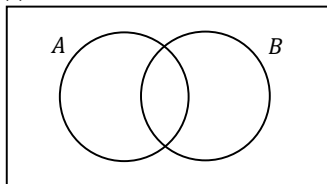
(a) $A \cap B'$



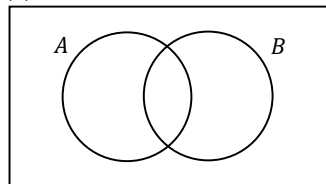
(b) $A' \cup B$



(c) $A' \cap B'$



(d) $A' \cup B'$



- 2 Given that $A \subset B$, simplify

(a) $A \cap B$

(b) $A \cup B$

(c) $A \cap B'$

(d) $A' \cup B$

- 3 Use the inclusion-exclusion principle to find the number of multiples of 2 or 3 in the set of integers from 1 to 100 inclusive.

- 4 The Head Master is feeling generous. He puts the 720 pupils in School into a list, and awards the book 'Twelfth Night' to every twelfth pupil. Additionally, he awards the book 'Catch-22' to every twenty-second pupil. How many pupils in total receive at least one book from the Head Master?

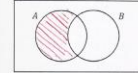
- 5 (a) How many different arrangements of the letters PATHOGEN are there?
 (b) How many different arrangements of the letters PATHOGEN are there in which the three vowels are together?
- 6 (a) How many different arrangements of the letters REPEATED are there?
 (b) How many different arrangements of the letters REPEATED are there in which the four consonants are together?

Commented [NS(1):

region indicated

$$\text{So } (A \cap B')' = A' \cup B$$

(a) $A \cap B'$



(b) $A' \cup B$



(c) $A' \cap B'$



(d) $A' \cup B'$



$$\text{So } A' \cap B' = (A \cup B)'$$

$$\text{So } A' \cup B' = (A \cap B)'$$

Commented [NS(2): A

Commented [NS(3): B

Commented [NS(4): Empty set

Commented [NS(5): Universal set

Commented [NS(6): There are $100/2=50$ multiples of 2. There are $100/3$ round down = 33 multiples of 3. There are $100/6$ round down = 16 multiples of 2 and 3 (i.e. multiples of 6). In total there are $50+33-16=67$ multiples of 2 or 3.

Commented [NS(7): The multiples of 12 and 22 are the multiples of 132. Now proceed as in Question 3 to get the answer $60+32-5=87$ pupils

Commented [NS(8): $8!=40,320$

Commented [NS(9): Treat the three vowels as a single block that can take $3!$ different forms. There are 6 blocks to arrange, so $6!*3! = 4320$

Commented [NS(10): $8!/3! = 6720$

Commented [NS(11): Put the four consonants into a single block which has $4!$ different forms. There are now five blocks to rearrange (three of which are identical), so $4!*5!/3!=480$

- 7 How many five-digit numbers are there in which the digits are in strictly increasing order?
- 8 How many six-digit even numbers can be made by rearranging the numbers 012345?
- 9 (a) How many different selections of five letters from MACHINE are there (the order in which the letters are selected doesn't matter)?
- (b) How many different selections of five letters from MACHINE are there which have exactly two consonants?
- 10 A committee of seven people needs to be chosen from a group of ten women and eight men.
- (a) How many different possible committees are there?
- (b) How many committees are there that consist of four women and three men?
- (c) How many committees are there that have at least one woman and at least one man?
- 11 (a) How many ways are there of splitting a group of nine people into two teams of sizes four and five?
- (b) How many ways are there of splitting a group of nine people into three teams each of size three?
- 12 How many different arrangements of the letters COFFEE are there in which there is no repeated letter? (That is, the two Es cannot be together, and the two Fs cannot be together.)
- 13 (a) How many different patterns can be made when two blue balls, two red balls, a yellow ball, and a green ball are put in a line?
- (b) How many different patterns can be made when two blue balls, two red balls, a yellow ball, and a green ball are put in a line if no balls of the same colour are placed next to each other?
- 14 A triangle has side length 2 cm, and five points are chosen inside the triangle. Use the pigeonhole principle to show that at least two of these points are no more than 1 cm apart.
- 15 Use the pigeonhole principle to show that for every positive integer n there exists a number divisible by n that consists only of the digits 0 and 5.
[Hint: Consider the sequence 5, 55, 555, 5555,]
- 16* Jon has seven cards with the numbers 0, 0, 2, 3, 4, 5, 7 on them. How many different five-digit even numbers can be made by using five of Jon's cards?

Commented [NS(12): There are hard ways to approach this but hopefully while exploring those hard ways, you'll realise that once you know what the five digits are, you have no freedom over their order. Therefore we only need to choose the five digits. We can't choose 0 (because then the number will have to start 0) so $9C5=126$

Commented [NS(13): Two approaches:

Three choices of starting value and $5!$ ways to rearrange the rest, then subtract the ones that start zero: $3*5!-2*4!=312$

Ending zero + not ending zero: $5!+2*4*4!=312$

Commented [NS(14): $7C5=21$

Commented [NS(15): Must have all three vowels and 2 of the four consonants. So $4C2=6$

Commented [NS(16): $18C7=31,824$

Commented [NS(17): $10C4 * 8C3 = 11,760$

Commented [NS(18): $10C7 + 8C7 = 128$ committees that are all female or all male. So $31824-128 = 31,696$

Commented [NS(19): $9C4 = 126$

Commented [NS(20): $9C3 * 6C3 / 3! = 280$

Commented [NS(21): Treat the EE as a single block: $5!/2! = 60$

Treat the FF as a single block: 60

Treat EE and FF as single blocks: $4! = 24$.

So $60+60-24=96$ arrangements that have at least one repeated letter.

Subtract from the total number of arrangements: $6!/2!2! - 96 = 84$.

I can't see a nice way of doing this without inclusion-exclusion.

Commented [NS(22): $6!/2!2!=180$

Commented [NS(23): This question is 'isomorphic' to Question 12. So the answer is once again **84**

Commented [NS(24): Choosing five points gives a hint to split the triangle up into four parts. Do this by joining the midpoints of the sides of triangle with straight lines. Let the four triangles thus created be the pigeonholes and let the five points be the pigeons. So at least one of the four little triangles contains (at least) two points in or on it. These points are then no more than 1 cm apart.

Commented [NS(25): Without the hint, this would be far too difficult for an exam, but with the hint it is challenging but do-able. The key is to realise that the differences between the numbers in the sequence will consist of only the digits 0 and 5. So we just have to prove that there exist two elements from the sequence whose difference is divisible by n . This is now a standard pigeonhole argument: Consider the remainders when each of the numbers in the sequence is divided by n . There are only n possible remainders (0 to $n-1$) so in any collection of $n+1$ numbers there must be at least two which have the same remainder when divided by n . The difference between these numbers will then be divisible by n .

Commented [NS(26): This question is too hard for an exam. Any of the individual cases that arise might be suitable, however. The easiest approach I've found is to partition into the following cases: No zeros, ends zero, 1 zero and doesn't end zero, 2 zeros and doesn't end zero.

$2*4! + 5*5*4*3 + 2*4*3C2*3! + 2*4*3C1*3 = 564$